Application of Curiosity Driven Exploration Algorithms for Hardware Interference Identification

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Ludovic Matar | 16 octobre 2026



Inria – Centre Inria de l'Université de Bordeaux

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Outline

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Context

- The shift from single-core to multi-core architectures is essential in safety-critical embedded systems in multiple domains.
- Introduces new complexities due to tasks running in parallel competing for shared resources ; inter-core interference.
- AMC 20-193 standard requires identifying all sources of interference that can occur when executing tasks in parallel and addressing them.

Purpose

Demonstrate the ability to build a map from complex system behaviors to their causes.

- Great diversity of interference features involving shared combinations of shared resources by using available detailled information from the simulated platform.
- With limited experimental budget, more behaviors than sampling entries of the simulator randomly.
- Guidance for analyzing the results.

Interference sources

Definition

We call a "pair of programs" a couple of two programs of code S_1 and S_2 allocated respectively to the C_1 and C_2 cores of an execution platform. S_1 and S_2 are independent code programs. In particular, they do not exchange data.

- **Cache contention :** multiple cores compete for space in shared caches, evicting each other's data and causing cache misses.
- **Memory contention :** multiple cores access a shared memory simultaneously (such as the main memory), leading to a bottleneck due to limited memory bandwidth.
- **Bus contention :** cores compete for access to a shared bus, resulting in bandwidth limitations and delays in data transfer.
- **I/O Contention :** multiple processes try to perform I/O operations at the same time, leading to delays and reduced I/O performance.

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Our simplified simulator

Our simulator model contains the following **resources** :

- two cores,
- their respective private L1 caches,
- a shared L2 cache unit
- DDR
- DDR controller,
- an interconnect model.

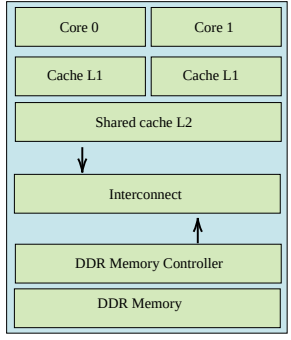


Figure – Simulated dual-core architecture used in this work.

How the simulator works

- Takes simplified assembly memory instructions
- Distinct sets of addresses for the cores, i.e., core 0 : 0,9 and core 1 : 10,19.

```
1 program_core_0 = {23:('read', 11),37:('read',17),49:('write',6)}
2 program_core_1 = {12:('write', 24),18:('read',39),32:('write',37)}
```

- Outputs description of interference events for all resources.

Resources	Description	Features
L2 cache interference	Cross-core eviction tracking	set_idx, evicted_core_id, evicted_instr_id, evicted_addr, causing_core_id, causing_addr
BUS contention	Delayed requests due to bus bandwidth limitations	nb_competing_requests
DDR interference	Delayed requests due to bank conflicts	row,bank,addr,req_type,scheduled_delay,c ore_id
DDR scheduler interference	Delayed requests due to higher priority requests	bank,addr,req_type,scheduled_delay,core_id, core_attacker,type_attacker, addr_attacker,bank_attacker,row_attacker

Table – Events tracked by the simulator

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Overview of the method

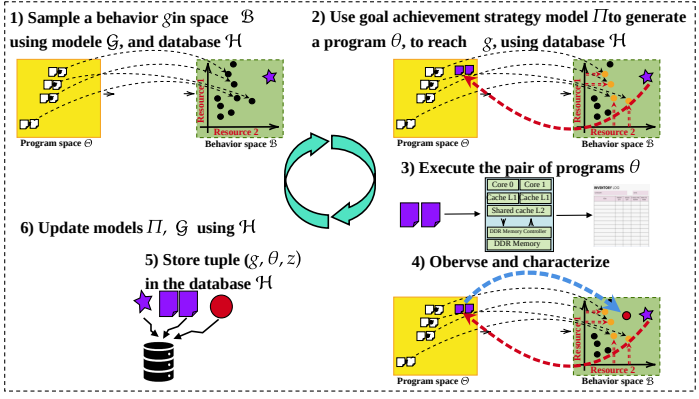


Figure – IMGEP for our case of interference identification, based on the general description of IMGEP

Behavior space and goal sampling

- We aim at covering the behavior space $\mathcal{B}$ of interference on the architecture.
- Behavior space $\mathcal{B}$ made from several parts $\mathcal{B}_r$ for each resource $r : \mathcal{B} = \cup_r^n \mathcal{B}_r$. Events for resources r are stored in part $\mathcal{H}[r]$ of $\mathcal{H}$.
- One wants to discover as many interference as possible among all the possible combinations of resources,
- One chooses to bias the discovery process by sampling goals.

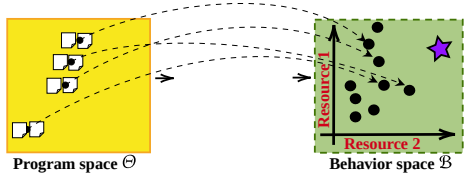


Figure – The purple star represents the goal. The axis indicate the agent tries to reach an experiment outcome with at least one interference for each required resource.

Goal sampling in two steps

What combination a^* to choose ?

- $\mathcal{P}(\mathcal{C})$, powerset of $\mathcal{C}$ ={**DDR, DDR scheduler, Cache, Bus**} contains $2^4 - 1 = 15$ combinations.
- Example : a_4 means that at least an interference on the L2 cache is found, but none for the other resources.
- Random variable g of support $\mathcal{S} = \cup_j^{card(\mathcal{P}(\mathcal{C}))}\{a_j^*\}$ determines what goal the agent chooses.
- Needs to find an optimal law of g $\forall a^* \in \mathcal{S}, \mathbb{P}_{\mathcal{H},g}(\{a^*\}) = \mathbb{P}_{\mathcal{H}}(g = a^*) \in [0, 1]$

Table – resource combinations

ref.	BUS	DDR	DDR Scheduler	L2 Cache
a_1	•	×	×	×
a_2	×	•	×	×
a_3	×	×	•	×
a_4	×	×	×	•
a_5	•	•	×	×
a_6	•	×	•	×
a_7	×	•	×	•
a_8	×	•	•	×
a_9	×	•	×	•
a_{10}	•	•	•	×
a_{11}	•	•	×	•
a_{12}	•	•	×	•
a_{13}	•	×	•	•
a_{14}	×	•	•	•
a_{15}	•	•	•	•

Table – resource combinations

Goal sampling in two steps

What combination a^* to choose ?

- Uniform sampling over $\mathcal{S}$:
  $\mathbb{P}_{\mathcal{H},g}(\{a^*\}) = \frac{1}{card(\mathcal{S})}, \forall a^* \in \mathcal{S}$
- Sampling based on exploration difficulty :
  $\mathbb{P}_{\mathcal{H},g}(\{a^*\}) = \frac{exp(-n_{\mathcal{H}}(a))}{\sum_j exp(-n_{\mathcal{H}}(a_j))}, \forall a^* \in \mathcal{S}$
- Diversity progress based sampling :$\mathbb{P}_{\mathcal{H},g}(\{a^*\}) = \frac{n_{\mathcal{H}}(a)}{\sum_j n_{\mathcal{H}}(a_j)}, \forall a^* \in \mathcal{S}$

Table – resource combinations

ref.	BUS	DDR	DDR Scheduler	L2 Cache
a_1	•	×	×	×
a_2	×	•	×	×
a_3	×	×	•	×
a_4	×	×	×	•
a_5	•	•	×	×
a_6	•	×	•	×
a_7	×	•	×	•
a_8	×	•	•	×
a_9	×	•	×	•
a_{10}	•	•	•	×
a_{11}	•	•	×	•
a_{12}	•	•	×	•
a_{13}	•	×	•	•
a_{14}	×	•	•	•
a_{15}	•	•	•	•

Table – resource combinations

Goal sampling in two steps

One samples vectors for the each of the resource of the combination.

- Firstly, the agent samples a^* using the law $\mathbb{P}_{\mathcal{H},g}$ of g.
- Secondly, it samples vectors g_r associated to the resources or distinct type events $r \in a$.
- g_r are the furthest from 5 sampled candidates in order to maximize novelty in discoveries.
- Sampling formula :

$$g_r \sim \mathcal{U}([f_1 \cdot (min_{\mathcal{B}_r} z)_1, f_2 \cdot (min_{\mathcal{B}_r} z)_2]) \otimes \cdots \otimes \mathcal{U}([f_1 \cdot (min_{\mathcal{B}_r} z)_{\dim(\mathcal{B}_r)}, f_2 \cdot (max_{\mathcal{B}_r} z)_{\dim(\mathcal{B}_r)}])$$

Goal achievement strategy

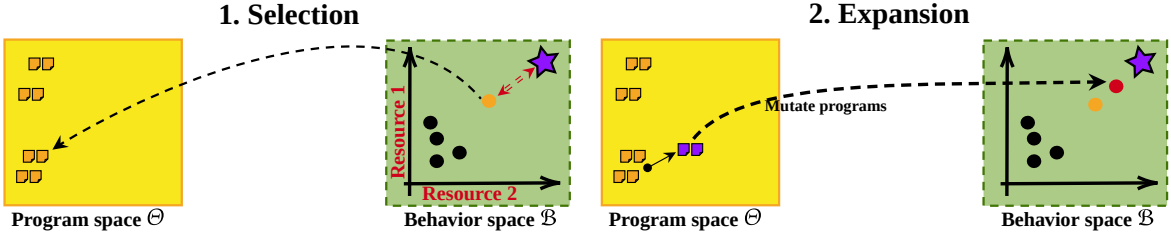


Figure – One selects the closest outcome (yellow)from the database $\mathcal{H}$, and one expands the history by producing a candidate parameter (purple)

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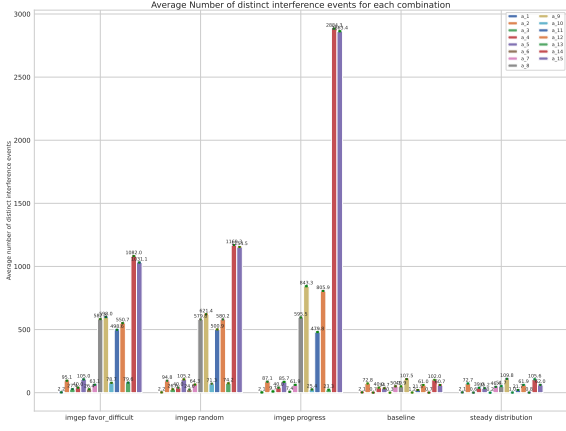
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Results

Number of distinct events for each resource combinations. This results from an example run of $N = 2 \cdot 10^4$ IMGEP iterations and a warming set of $N_{init} = 1000$ iterations of random program synthesis, and 1 random mutation from the mutation operator.



- IMGEP succeeds in finding multiple interference cases for every resource combination.
- As opposed to other steady sampling distributions.

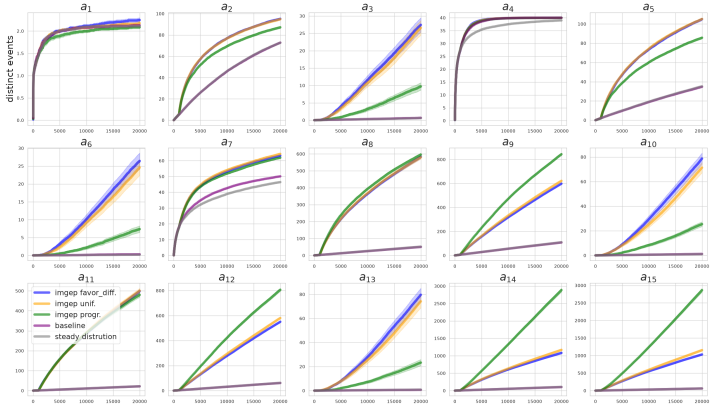


Figure – Evolution over the exploration of the number of distinct events for the different combinations of interference types.

Analysis of the dataset

- Most encountered features for $a \in \mathcal{S}$?
- $\implies$ Analyze Distribution of vectors $\{r : z_r \in \mathcal{B}(r) | \forall r \in a\}$.
- Too many features to be easily analyzed. $\implies$ Keep only the most relevant features $\implies$ Observe their distribution.

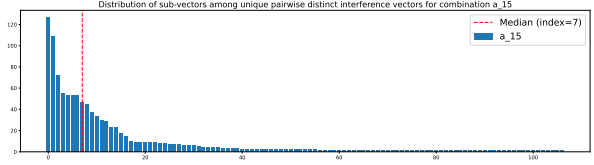


Figure – Among the set of pairwise distinct vectors for combination $a_{15} \in \mathcal{S}$, distribution of the subvectors featuring a selection of values. 7 distinct vectors responsible for 50 pourcents of a_{15} cases.

Analysis of the dataset

- Most encountered features for $a \in \mathcal{S}$?
- $\implies$ Analyze Distribution of vectors $\{r : z_r \in \mathcal{B}(r) | \forall r \in a\}$.
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	DDR :bank	DDR sched. :bank	DDR sched. :addr	DDR sched. :bank attacker	L2 cache :evicted core id	L2 cache :causing core id
1	5	5	13	5	1	0
2	2	2	2	2	1	0
3	5	5	13	5	0	1
4	4	4	4	4	1	0
5	1	1	9	1	1	0
6	5	5	5	5	1	0
7	5	5	5	5	0	1

Table – Most frequent (50 percent) vectors of features inducing a_{15}. Only a few features considered $\implies$ tracktable analysis.

Analysis of the dataset

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- $\implies$ Analyze Distribution of vectors $\{r : z_r \in \mathcal{B}(r) | \forall r \in a\}$.
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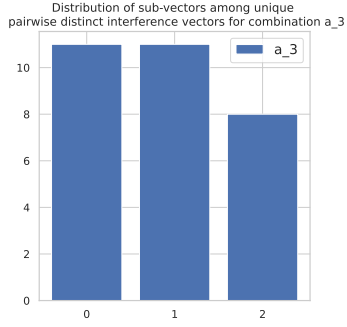


Figure – Among the set of pairwise distinct vectors for combination $a_3 \in \mathcal{S}$, distribution of the subvectors featuring a selection of values.

Analysis of the dataset

- Most encountered features for $a \in \mathcal{S}$?
- $\implies$ Analyze Distribution of vectors $\{r : z_r \in \mathcal{B}(r) | \forall r \in a\}$.
- Too many features to be easily analyzed. $\implies$ Keep only the most relevant features $\implies$ Observe their distribution.

bank	addr	bank attacker	row attacker
0	16	1	5
4	4	0	3
4	4	1	1

Timing data

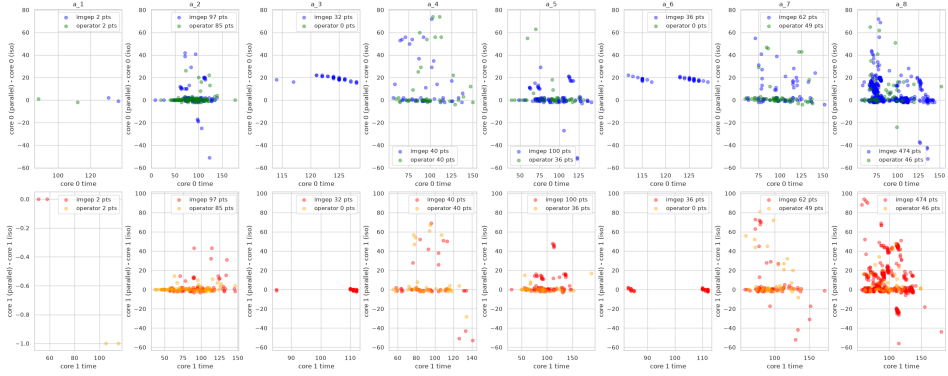


Figure – One observes that extending time does not determine the source of interference

Timing data

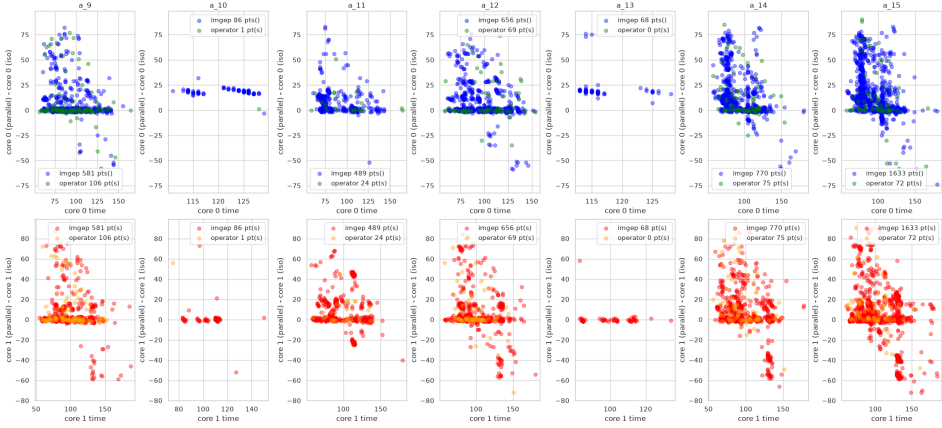


Figure – One observes that extending time does not determine the source of interference

Limitations

- The system could be more realistic, and a transition on a high fidelity simulator like gem5 could benefit the audit of this project.
- stochasticity of the results.

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Conclusion

- IMGEP relies on simple machine learning models to structure an autonomous agent that fixes his own goals, and on the establishment of a behavior and a parameter space
- Automated discovery algorithms can successfully lead to the identification of complex system behavior causes.
- Encourage the idea of exploring more complex simulated platfoms reproducing real multicore architectures